

HLD

PW SKILLS

HIGH LEVEL DESIGN (HLD)

Project:

Cryptocurrency Historical Prices Prediction using Machine Learning

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Introduction

The High-Level Design (HLD) document provides an overview of the Cryptocurrency Historical Prices Prediction system. It explains the overall architecture, major components, data flow, and technologies used in the project. The system is designed to predict cryptocurrency market volatility using machine learning techniques.

Project Objective

The objective of this project is to develop a machine learning model capable of predicting cryptocurrency volatility based on historical market data. The project aims to assist investors and traders in understanding future market fluctuations using historical OHLC prices, trading volume, and market capitalization.

System Architecture

The system consists of the following major modules:

- Data Collection
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Data Scaling
- Model Training
- Model Evaluation
- Prediction Generation
- Model Saving

Technologies Used

Programming Language:

Python

Development Environment:

Google Colab

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib

Dataset:

Cryptocurrency Historical Prices Dataset

Input

Input:

Historical cryptocurrency market dataset containing:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Market Capitalization
- Date
- Cryptocurrency Name

Output

Output:

- Predicted Cryptocurrency Volatility
- Performance Metrics
- Saved Machine Learning Model
- Prediction CSV File

Advantages

- Fast prediction of market volatility.
- Automated preprocessing pipeline.
- Accurate prediction using Random Forest Regressor.
- Easy deployment for future applications.
- Scalable architecture for larger datasets.

Conclusion

The High-Level Design provides a clear overview of the project architecture and workflow. It defines the major modules and demonstrates how raw cryptocurrency data is transformed into meaningful volatility predictions using machine learning techniques.